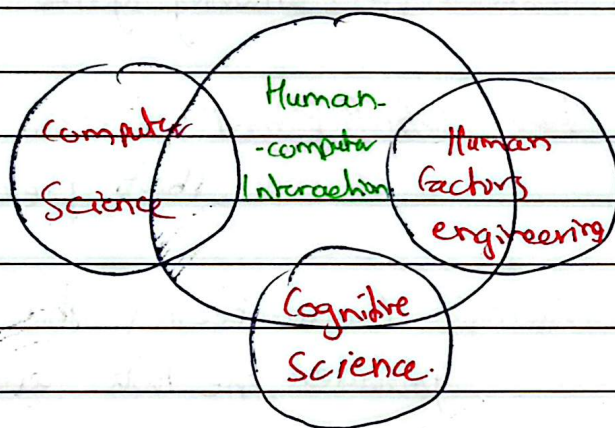


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Intro to HCI

Human Computer Interaction is a multi-disciplinary field of study, focusing on the design of computer technology.



Multidisciplinary field.

- It focuses on how humans interact with computational technologies & how to design, evaluate and implement interactive systems that safe & intuitive to use.

Goals of HCI:

The **ultimate goal** of HCI is to improve the relationship between humans & machines by optimizing system design around human capabilities, needs & limitations.

- 1) **Usability**: Ensuring systems are easy to learn.
- 2) **User Safety & Error Minimization**: Designing interfaces that prevent user errors.
- 3) **Utility**: Guaranteeing the software actually provides the tools and features users need.
- 4) **Positive UX**: Creating interfaces that are engaging, accessible & satisfying.
- 5) **Accessibility & Inclusivity**: Designing software so that all users, even disabled ones can use it.

Importance of HCI:

It can affect:

- effectiveness
- Productivity
- Morale
- Safety - designing

Variations in HCI

All users are different so designers must account for human variations across sensory, mental & social dimensions.

Human Factor	HCI	key design points
1) Human variations:	Differences in age, physical ability, culture & language	• provide multiple language options
2) Perceptual phenomenon	How humans sense & interpret visual, auditory & tactile input from screen	• Use strong color contrast, clear visual hierarchies & apply Gestalt Principles

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3) Cognitive

How users process information, pay attention, reason & store memory.

Minimize cognitive load. e.g by avoiding overly complex menus.

4) Emotional.

The user's affective state (Joy, grief, anxiety)

Provide clear error messages & aesthetically layouts.

5) Motor

Physical Movement capabilities, fine motor control, hand-eye coordination.

Design larger click/Tap targets for small screens or ageing users.

6) Social

How technology facilitates or alters communication & group dynamics.

Build clear indicators for presence & obvious privacy controls for sharing.

7) Behavioral / Psychological.

Habitual actions, interaction routines, user expectations.

Stick to familiar UI patterns so users don't have to learn basic navigation again

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4 principles of scalable, user-friendly softwares:

1) Standardization

Adhering to established industry guidelines, rules & hardware/software norms across systems.

Benefit → Lowers the learning curve. Users rely on existing mental models instead of relearning how to navigate.

Example → Keyboard shortcuts like Ctrl+C & Ctrl+V ~~are considered~~ ~~are~~ have same copy/paste across multiple OS.

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2) Integration:

Connecting distinct software components, tools or services so they exchange data & work seamlessly.

Benefit → Saves user effort by eliminating repetitive data entry & switching friction between apps.

Example → Clicking a web link inside a messaging app to open it seamlessly in browser without interrupting your flow.

3) Consistency:

Keeping design elements, terminology, layouts & behaviours identical across the entire application or ecosystem.

Benefit → Prevents user confusion, reduces cognitive load & speeds up task execution.

Example → Using the exact same red color, icon style & modal prompt for every delete action across the app.

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4) Portability:

The capability of a software or user interface to run effectively across different hardware platforms, OS versions or screen sizes.

Benefit → Ensures uniform access, flexibility & operational continuity regardless of device constraints.

example → Web apps like Figma, canva run identically on windows, macOS or linux.

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AU 4:

• Consistency vs Standardization:

Internal

making app look & behave same across all its pages.

external

making app conform to universal industry rules.

• Integration + portability:

portable apps mostly rely on integrated ~~cloud~~ cloud backend APIs so users can start a task on a phone & finish it seamlessly on a desktop.

• Example (general):

Norman's door:

Don Norman said that if a door has a handle it should be a pull & with handle should be a push. This saves us from being confused & makes a door easy to use.

Usability measures:

How can we measure goodness of an ~~interface~~ interface?

A few metrics have been defined to evaluate an interface:

Some examples are:

- 1) Task Success Rate (Effectiveness)
- 2) Time on Task (Efficiency)
- 3) Error Rate
- 4) System Usability Scale
- 5) Single Ease Question
- 6) User feedback

Official standard for usability (ISO-9241).

This organizes all metrics into 3 pillars:

- 1) Effectiveness
- 2) Efficiency
- 3) Satisfaction

Software engineering VS HCI:

Software Engineering	HCI
• System centered • Software architecture • Procedural aspects.	• User centered • Users • Tasks • Context of Use

Machine- and people-centred views

View	People are	Machines are
Machine-centred	Vague Disorganised Distractible Emotional Illogical	Precise Orderly Undistractible Unemotional Logical
People-centred	Creative Compliant Attentive to change Resourceful Able to make flexible decisions based on content	Dumb Rigid Insensitive to change Unimaginative Constrained to make consistent decisions

HCI is mainly people centered but tries to bring a balanced trade off between both.